DATASET SELECTION BRIEF

Flickr-8k Audio Caption Corpus

Finding Images Through Spoken Language — A Contrastive Learning Approach

Purpose. This brief records why this dataset is a promising candidate, what its files contain, how it may support our audio-to-image retrieval project, and what must be checked before use.

1. Dataset link

Kaggle page: https://www.kaggle.com/datasets/warcoder/flickr-8k-audio-caption-corpus
Check the page directly for the current file list, download size, license, and usage conditions.

2. Why we are using / considering this dataset

The project aims to retrieve relevant images from spoken-language queries. This corpus is a strong candidate because its description reports 40,000 spoken English captions in WAV format, linked to captions and images from the Flickr8k corpus. It gives us real human speech to investigate instead of requiring us to generate all audio with text-to-speech (TTS).

Main benefit: real recorded speech is relevant to a system that accepts spoken queries, and the speaker mapping can support speaker-aware analysis.

Important: the provided description says the audio and mapping files are included, but the matching image files and caption-text file may need to be downloaded separately from the original Flickr8k corpus.

3. Dataset facts from the supplied description

Item	Description
Audio	40,000 spoken caption recordings in .wav format.
Sample rate	16,000 Hz (16 kHz).
Bit depth / format	16-bit audio in Microsoft WAVE format.
Speakers	183 unique speakers, identified in wav2spk.txt.
wav2capt.txt	Maps each WAV filename to an image filename and caption number.
wav2spk.txt	Maps each WAV filename to a speaker ID.
Flickr8k.token.txt	From the original corpus; maps image filenames and caption numbers to written caption text.
Splits	Description references Flickr_8k.trainImages.txt, Flickr_8k.devImages.txt, and Flickr_8k.testImages.txt.

4. How it fits the project

For each spoken recording, the mapping file identifies the image and caption number. The original caption file provides the text. We can then build aligned examples containing audio, its caption, and its matching image.

For contrastive learning, an audio encoder and an image encoder (possibly with projection layers) must be trained or adapted to produce compatible representations. Matching audio-image pairs are encouraged to be close in the shared embedding space; mismatched pairs are used as negatives. This dataset provides aligned examples, not a trained model.

5. Benefits and risks to check

Potential benefits	Checks / limitations
• Real human speech rather than only synthetic TTS. • 40,000 recordings for experiments. • Standard 16 kHz WAV format. • Speaker IDs can support speaker-aware evaluation.	• Matching images and caption text may be separate downloads. • Verify mappings against the exact Flickr8k release. • Check license and permitted academic use. • Inspect duration, audio quality, missing/corrupt files, and speaker distribution. • Avoid image overlap across train/dev/test that could cause leakage.

6. Recommended next steps

1. Confirm team approval and record the dataset source/version.
2. Check Kaggle's file list, download size, license, and usage terms.
3. Download and extract the audio corpus without renaming original files.
4. Obtain matching Flickr8k images and caption text if absent.
5. Inspect a small sample in Python: WAV readability, duration, mapping, caption, and image.
6. Check missing files and duplicates, then preserve the original train/dev/test split.
7. Build the audio preprocessing and embedding pipeline.

7. Current decision status

Status: selected as the team's candidate for initial exploration, pending final checks.
Not yet verified: exact package contents, license/usage permissions, image availability, or whether the dataset meets all project needs.

Basis of this brief: the Kaggle dataset description supplied in the project conversation and the dataset page linked above. Reported details should be verified against the downloaded release.